Reversible Reactions

Exam code: 0620 & 0971

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Reversible reactions

- Some reactions go to **completion**, where the reactants are used up to form the product molecules and the reaction stops when all of the reactants are used up
- In **reversible reactions**, the product molecules can themselves react with each other or decompose and form the reactant molecules again
- It is said that the reaction can occur in **both directions**:
  - The **forward** reaction forming the products
  - The **reverse** reaction forming the reactants
- If the forward reaction is exothermic, then the reverse reaction will be endothermic
  - The same amount of heat is transferred in both directions

Help me understand

Chemical equations for reversible reactions

- When writing chemical equations for reversible reactions, the following symbol is used:

$$\rightleftharpoons$$

- An example is, the reaction for the Haber process which produces ammonia from nitrogen and hydrogen

$$N_2 + 3H_2 \rightleftharpoons 2NH_3$$

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Ask questions and get help with exactly what you're revising.
Got it

Hydrated and anhydrous salts

- **Hydrated** salts are salts that contain **water of crystallisation** which affects their molecular shape and colour
  - Water of crystallisation is the water that is stoichiometrically included in the structure of some salts during the crystallisation process

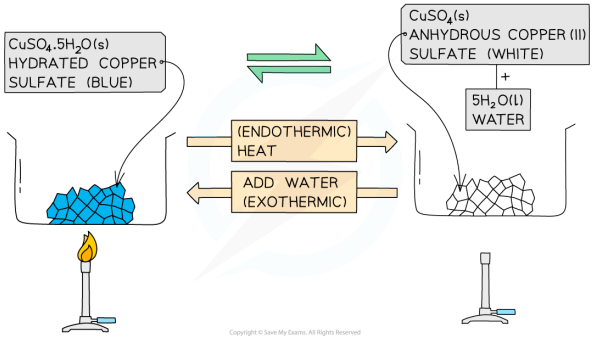
- One example is copper(II) sulfate:

hydrated copper(II) sulfate ⇌ anhydrous copper(II) sulfate + water

$$CuSO_4 \bullet 5H_2O \rightleftharpoons CuSO_4 + 5H_2O$$

- The hydrated salt is copper(II) sulfate pentahydrate, $CuSO_4 \bullet 5H_2O$
  - These are usually seen as blue crystals
  - The hydrated salt can be heated / dehydrated to form anhydrous copper(II) sulfate, $CuSO_4$
  - This reaction is endothermic as energy is taken in to remove the water

- The anhydrous salt is copper(II) sulfate
  - This is usually seen as white crystals / powder
  - Adding water to the anhydrous salt forms the hydrated copper(II) sulfate pentahydrate, $CuSO_4 \bullet 5H_2O$
  - This reaction is highly exothermic



The forward reaction is exothermic and the reverse reaction is endothermic

- Another example is cobalt(II) chloride:

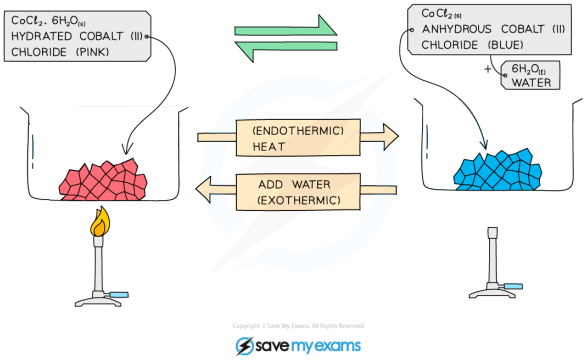
hydrated cobalt(II) chloride ⇌ anhydrous cobalt(II) chloride + water

$$CoCl_2 \bullet 6H_2O \rightleftharpoons CoCl_2 + 6H_2O$$

- The hydrated salt is cobalt(II) chloride hexahydrate, $CoCl_2 \bullet 6H_2O$
  - These are usually seen as pink crystals
  - The hydrated salt can be heated / dehydrated to form anhydrous cobalt(II) chloride, $CoCl_2$
  - This reaction is endothermic as energy is taken in to remove the water

- The anhydrous salt is cobalt(II) chloride, $CoCl_2$
  - This is usually seen as blue crystals

- This reaction is highly exothermic



The forward reaction is exothermic and the reverse reaction is endothermic

Examiner Tips and Tricks

The hydration of $CoCl_2$ and $CuSO_4$ are chemical tests used to detect the presence of water.

You should remember the equations and colour changes:

- $CoCl_2 + 6H_2O \rightleftharpoons CoCl_2.6H_2O$ **Blue to pink**
- $CuSO_4 + 5H_2O \rightleftharpoons CuSO_4.5H_2O$ **White to blue**

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